NAZIM HUSSAIN

AI & Machine Learning Engineer | LLMs, RAG & Computer Vision

Karachi, Sindh, Pakistan • +92 333 7006557 • nazimali8172@gmail.com • linkedin.com/in/nazim-hussainn • github.com/nazim-hussain1 • Portfolio

PROFESSIONAL SUMMARY

AI Engineer with a BS in Artificial Intelligence (CGPA 3.42/4.00) specializing in production-grade Large Language Model (LLM) architectures, hybrid Retrieval-Augmented Generation (RAG) pipelines, and real-time computer vision systems. Proven track record architecting end-to-end intelligent systems, including a bilingual legal RAG platform with sub-2-second retrieval and edge-optimized face verification with anti-spoofing liveness checks. Proficient in Python, PyTorch, LangChain, FAISS, and building scalable APIs with FastAPI.

TECHNICAL SKILLS

AI, LLMs & Retrieval: Large Language Models (LLMs), RAG, LangChain, FAISS, BM25, Cross-Encoder Re-Ranking, Semantic Search, Prompt Engineering
Computer Vision & ML: PyTorch, TensorFlow, OpenCV, InsightFace (ArcFace, SCRFD), CNNs, Scikit-learn, Feature Extraction
Backend & Systems: Python, FastAPI, Flask, SQLite, MySQL, REST APIs, NumPy, Pandas
Tools, Cloud & DevOps: Git, GitHub, Docker, Hugging Face Spaces, Google Colab, Linux/Bash, PyInstaller

FEATURED ENGINEERING PROJECTS

Pakistan Legal Advisor – Bilingual Legal RAG Platform | Python, LangChain, FAISS, BM25, Cross-Encoder **Live Demo**

- Architected an end-to-end bilingual (Urdu & English) legal retrieval platform indexing 500+ statutory legal documents to provide hallucination-resistant legal analysis.
- Engineered a hybrid retrieval engine combining dense vector indexing (FAISS) and sparse lexical retrieval (BM25) with cross-encoder re-ranking, achieving sub-2-second response latency.
- Containerized and deployed the complete inference and user query pipeline live on Hugging Face Spaces for public access.

Real-Time Face Recognition & Anti-Spoofing System | InsightFace, SCRFD, ArcFace, OpenCV **Notebook**

- Built an edge-compatible biometric verification pipeline combining SCRFD face detection with ArcFace 512-dimensional normalized feature embeddings.
- Implemented adaptive cosine similarity matching against a registered gallery and integrated a motion-based liveness check to block static photo and screen spoofing attacks.

Naz Poultry Manager – Enterprise Desktop Software | Python, CustomTkinter, SQLite, ReportLab, PyInstaller **GitHub**

- Engineered an offline-first desktop ERP software to automate commercial poultry operations, daily egg yields, and debtor balances with automated SQLite backup mechanisms.
- Built automated billing supporting 80mm ESC/POS thermal printing for point-of-sale transactions and dynamic A4 PDF statements via ReportLab; packaged as a zero-dependency .exe.

Handwritten Digit Classifier & Interactive Canvas | TensorFlow, Keras, OpenCV, Gradio, CNN **Notebook**

- Trained a deep CNN with Batch Normalization and Dropout on MNIST, paired with custom OpenCV preprocessing (Otsu thresholding, bounding-box centering) and interactive Gradio canvas.

Algorithmic Trading Signal Engine | Python, OpenCV, Quantitative Analysis **Prototype**

- Developed an automated trading signal generator using OpenCV contour and edge detection to capture market structure breaks, support/resistance zones, and volume spikes.

WORK EXPERIENCE

Computer Operator — *Citizen Police Liaison Committee (CPLC)* **Jan 2023 – Dec 2023 | Shaheed Benazirabad, Pakistan**

- Administered relational databases and digitized police-civilian records under strict security protocols with zero integrity breaches.
- Streamlined cross-departmental record verification and automated data entry workflows, reducing documentation turnaround time.
- Maintained hardware, operating systems, and local area network uptime to prevent administrative operational downtime.

EDUCATION & ACHIEVEMENTS

Bachelor of Science in Artificial Intelligence (BS AI) — *QUEST Nawabshah* **Dec 2022 – Dec 2026 | CGPA: 3.42 / 4.00**
Key Coursework: Deep Learning, Natural Language Processing (NLP), Computer Vision, Neural Networks, Database Systems.
Honors & Contributions: Final Year Project on Bilingual Legal RAG; Active Kaggle Contributor with 9 platform achievement badges.